

Propulsion System

Reference Performance System

Holybro doesn't publish a full thrust-curve datasheet for the S500 V2 2216-920KV motor, but it's the same physical motor family as T-Motor's Air Gear 2216 (900–920KV class) — historically what the S500 kit shipped with. The closest public bench-test data point, run on 4S with a 1045 prop .

50% throttle → ~430 g thrust at ~8.5 A per motor (4S / 14.8V)

I used this as the anchor point and modelled current vs. thrust with the standard propeller relation $\text{power} \propto \text{thrust}^{1.5}$ (since $\text{power} \propto \text{RPM}^3$ and $\text{thrust} \propto \text{RPM}^2$).

Component	Weight
Frame (lightweight 10" custom/ S500 style – not stock, too expensive)	220 g
4× Motors — Holybro S500 V2 2216-920KV (63 g each, per Holybro spec)	252 g
4× Propellers — Pro-Range 1045 (~12 g each)	48 g
4× ESC — 40A BLHeli_32 mini (~10 g each, see sizing below)	40 g
Pixhawk 6C Mini (39.2 g, per Holybro spec)	39 g
GPS + compass module (e.g. M10)	25 g
Power module (Holybro PM06, 24 g per spec)	24 g
ELRS receiver (nano)	5 g
Telemetry radio (SiK V3, 915MHz — lighter than RFD900x if weight is tight; swap in the RFD900x's ~45g if range matters more)	14-17 g
VTX + FPV cam (lightweight digital, e.g. Walksnail Moonlight)	30 g
Companion computer + camera (Raspberry Pi 5 8GB + OAK-D Lite)	126 g
Wiring, connectors, mounting hardware, standoffs	About 40 g
Subtotal of everything except battery	865 g

Sizing the Battery

The endurance formula:

Endurance (min) = (Battery capacity [mAh] × usable fraction) / (Total current draw [A] × 1000) × 60

Battery (4S) in mAh	Pack Weight	AUW	Hover current	Endurance	Margin under 2.5kg
5000	448 g	1.31 kg	24.3 A	9.9 min	1187 g
6000	538 g	1.40 kg	26.6 A	10.8 min	1097 g
7000	628 g	1.49 kg	29.1 A	11.6 min	1007g
8000	718 g	1.58 g	31.6 A	12.2 min	917 g

My pick: 8000 mAh Lipo, 25C+ discharge rating

This is the smallest standard commercial size that clears the 12-minute floor with a bit of margin (12.2 min), landing the aircraft at 1.58 kg all-up weight — nearly a full kilogram under the 2.5 kg cap. That spare weight budget isn't wasted headroom: it's room for an actual rescue payload (a small flotation marker or dye packet to drop on a confirmed sighting).

Sizing The ESC

Using the same thrust-current model at full throttle [from holybro] (max thrust for this motor class on 4S/1045 is commonly cited around 1000g):

Estimated max current per motor at full throttle ≈ 30 A

Pick: 40A (or 45A) BLHeli_32 4-in-1 ESC — e.g. Holybro Tekko32 F4 4-in-1

This happens to be a very common, widely-stocked ESC class, and BLHeli_32 with DShot support gives cleaner, more reliable signalling to ArduPilot/PX4 than legacy PWM/OneShot.